Biomedical Image Analysis: AI imaging Coding Case Study

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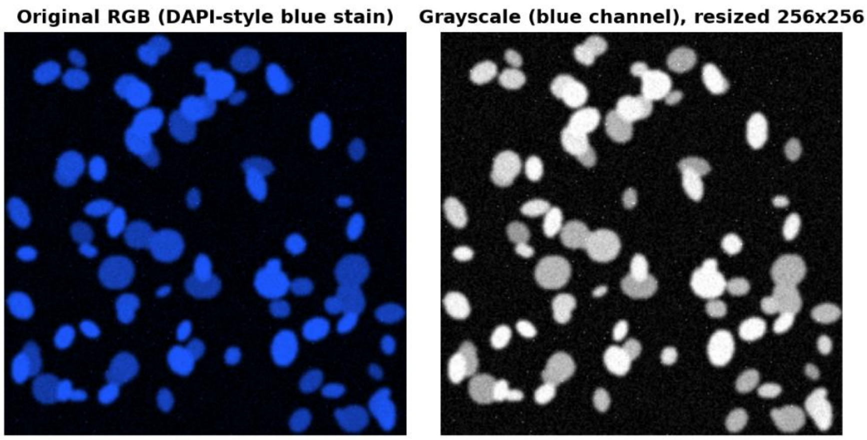
1. What the pipeline does

Every image passes through a processing pipeline consisting of four stages, and the main focus is on the development of reliable intermediate representations rather than on acceptance of some output as such. At stage one, the **llama3.2-vision** model processes the image in order to create a short JSON description. The second stage includes transformation of the same image using Otsu thresholding and connected components in order to obtain a table containing number of objects, their areas and shape properties. A special text-only model interprets this table without ever seeing the image itself. Stage three uses a U-Net model which is trained on 80 labelled images to provide a pixel mask. The fourth stage is similar to the third stage except that the U-Net mask is processed using the same numerical summariser from the second stage and then a local model transforms the output numbers into a short paragraph and JSON for each of the 12 test images. The reason behind this architecture is that all free-text outputs must be backed up with some verifiable intermediate representation.

Figure 1: Original RGB image and convert grayscale



2. Data preparation and describing it with a Vision Model

Figure 1 shows the RGB image together with its corresponding grayscale image. In the left panel, there are blue-stained nuclei on the image while the right panel shows the image after the conversion into single channel grayscale and resizing it to **256 × 256**. It can be seen that since the stain is mostly contained in the blue channel, the preprocessing step is done based on that channel instead of using the standard luminance weight on RGB. Since the blue channel only contributes about **11%** of the luminance according to standard luminance weights, this value is good for natural images but not appropriate here because most of the signals will be diminished.

Figure 2: Grayscale intensity histogram (*Top*) & representative images per density class (*Bottom*)

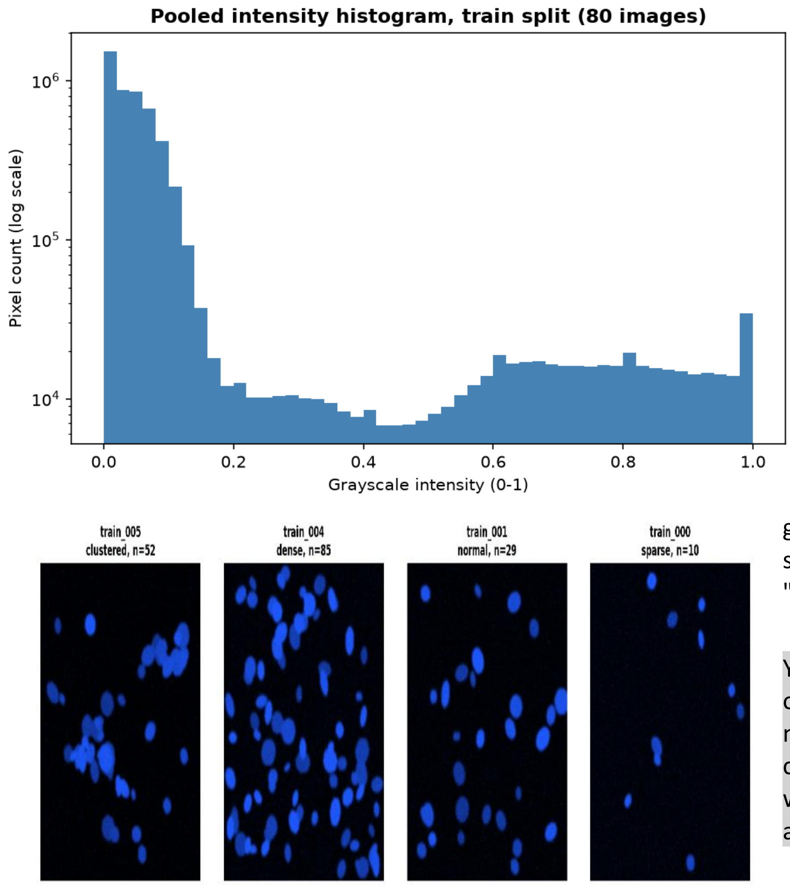


Figure 2 shows a training image for each density class, see ***bottom image*** (from right to left, they are sparse, normal, dense, and clustered) and the grayscale histogram of pooled 80 training images (***Top plot***), which were converted based on the method mentioned in ***Figure 1*** (the blue channel). There is a very large peak at **0** (the dark background, which makes up most of the area in the image, and another peak that is much less significant near **0.2** to **0.55**, which corresponds to nucleus pixels. The difference between the two makes it appropriate to use one global threshold Otsu.

The **engineered prompt** ensures that the model generates descriptions only, avoids any diagnosis, follows a specific JSON structure, and allows use of the word "uncertain" instead of making any guesses.

You are an assistant that gives purely **DESCRIPTIVE** observations about a microscopy image. You are NOT a medical professional and must NEVER provide a diagnosis, a clinical interpretation, or a disease name. Only describe what is visually present (shapes, colours, texture, spatial arrangement).

Look at the attached image and respond with ONLY a single JSON object (no extra text, no markdown code fences) with exactly these keys:

{"modality": "<your best guess >",

"tissue_type": "<what kind of biological structure is shown>",

"notable_features": "<one short sentence on notable visual features>",

"image_quality": "<one of: good, moderate, poor>"}

If you are not confident about a field, you MUST answer "uncertain" for that field rather than guessing. Do not include any text outside the JSON object.

Against a **naive prompt** (**'What do you see in this image? Please describe it.'**), the gap shows immediately. The naive answer read: *"The image appears to be a microscopic view of cells, likely under a fluorescence microscope. The blue dots are likely fluorescently labeled cells or cellular components, such as nuclei or mitochondria, that have been stained with a fluorescent dye. The image suggests that the cells are being studied for their morphology, structure, or function, possibly in the context of a biological or medical research project."* A guess as to the use of the image, which cannot be proven on the basis of the pixels in the image. The engineered prompt was generated three times with the temperature of 0.5 using the same image. Three generations did not converge into one explanation of what the image represents. **Run 1** saw the image as "blurry" and gave image_quality "poor", both of which are incorrect statements verifiable in *Figure 1.* **Run 2** identified the content as "fluorescence microscopy" and "cells" for modality and tissue type, which is a correct and concrete statement. **Run 3** gave "uncertain" for both fields. Three different confidence levels were produced in three runs with one being incorrect.

3. Classical features and a numbers-only description

The application of Otsu's algorithm on **train_004** (dense, **85** ground-truth nuclei) results in **45** connected components. In the sample of **80** training images, the correlation between the number of components and the ground truth reaches **r** = **0.83**, yet there may be up to **43** missed nuclei for the most difficult images (for example, **train_046** where **41** nuclei are detected, but there are **84** real ones) because of touching nuclei fusing in one object. The measurements of area and intensity are close to the ground truth (area error = **0.0005**; intensity error = **0.0023**), so the thresholding is accurate, but the only shortcoming is object counting.

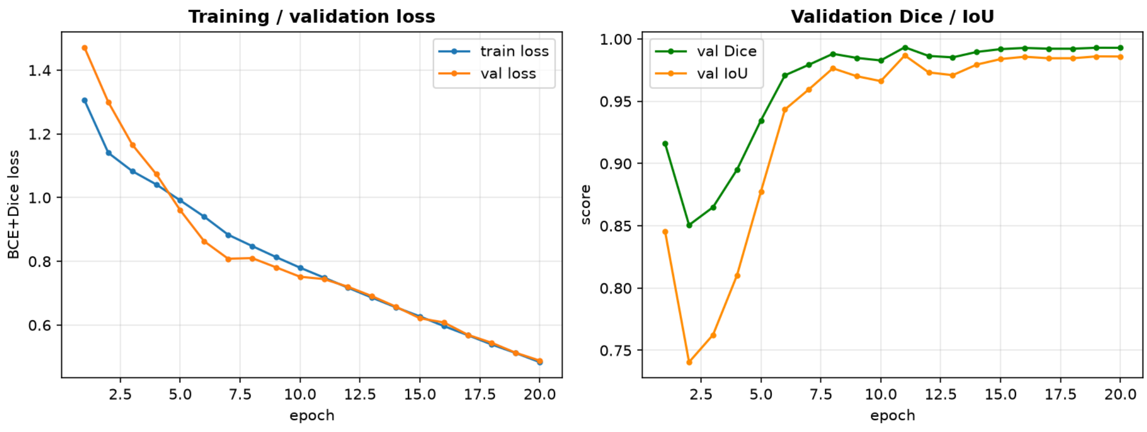
From the single string of statistics,"45 connected components...(dense). Object area: mean 293.0 **px^2...** solidity **0.92**... intensity 0.75" The text-only model interpreted that "**The dense object density and high solidity indicate that most objects are well-defined and not overlapping or merged. However, the irregular shape regularity and moderate quality flag hint at some inconsistencies"**

The sentence of the model claims the presence of "well-defined and not overlapping" objects, but its JSON report states "shape_regularity": "irregular," thus creating an inconsistency due to the same input data. In addition, the ending of the JSON output piece is missing; it is cut off at "quality_flag": "moderate," and therefore the notebook parser returned None.

4. Training the U-Net

The architecture, is the small U-Net from Lab 4 (consisting of three layers in encoder, one bottleneck, three layers in decoder, skip connections, and roughly **483,000** parameters) was trained on the actual training dataset of **80** images, and not on the simulated dataset of the laboratory. Training used a combined binary cross-entropy (**BCE**) and **Dice loss**, using Adam optimizer with a learning rate of **1×10^{-3}**, a batch size of **8**, and for **20** epochs.

Figure 4: training/validation loss and validation Dice/IoU curve plots



The results of ***Figure 3*** demonstrate smooth concurrent reductions in losses in both training and validation, without any signs of overfitting due to the limited size of the dataset. Moreover, the Dice coefficient demonstrates only one sudden rise in performance from about **0.92** to **0.99** within the first 10 epochs, after which it tends to be stable. This trend is consistent with a network switching from background classification to nucleus contour detection.

Table 1: Final Dice and IoU, Worst images

Metric	Value
Final validation mean Dice	**0.9929**
Final validation mean IoU	**0.9859**
Worst validation images by Dice	val_006.png = **0.9890** val_012.png = **0.9898** val_001.png = **0.9902**

Figure 5: input, ground truth, u-net prediction

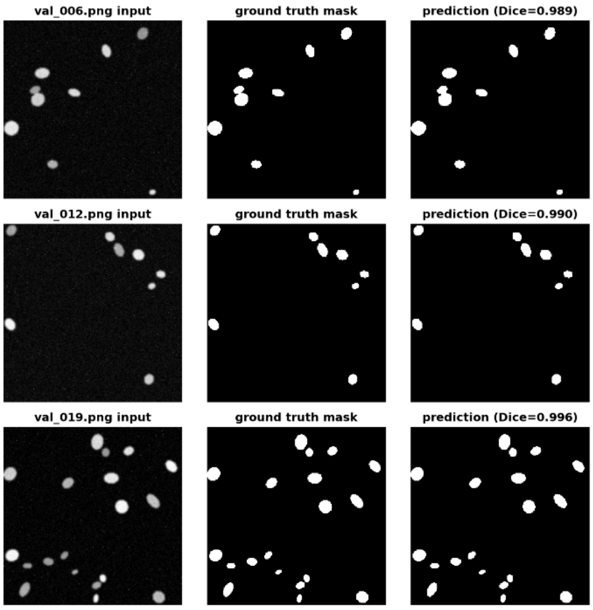


Figure 5 shows the input image, the ground truth, and the prediction of the two worst and one best scoring validation images. Even **val_006**, which is the worst among all twenty, covers its ground truth almost entirely both the tiny nucleus on the lower right-hand side and the two nuclei on the upper left-hand side that touch each other are correctly predicted.

5. Loss comparison, the hybrid pipeline,

and three extensions

Table 2 and ***Figure 6*** show how three different loss functions behave when using the same architecture and seed along with the same number of epochs, which is 10 in this case. The loss function with only **Dice** (*left plot*) demonstrates a sudden drop during training, as expected for such a loss function since it tends to produce vanishing gradients in the case when the prediction is close to zero. The plot on the right-hand side (*loss validation curves*) demonstrates the optimization process for each loss function. **BCE** has the best, most stable loss curve and achieves the highest final **Dice** and **IoU** scores in *Table 2*; however, taking into account the fact that there is quite a severe class imbalance in the dataset with a significant background class, it is still vulnerable to the above imbalance problem. The combined **BCE+Dice** loss sits between the two: it avoids the mid-training collapse seen with Dice and reduces the imbalance sensitivity of plain BCE, giving a more stable overall trajectory.

Table 2: loss ablation summary
Figure 6: Validation Dice / loss curves

Loss	Final val Dice	Final val IoU
BCE	0.9913	0.9828
Dice	0.9790	0.9589
BCE+Dice	0.9828	0.9662

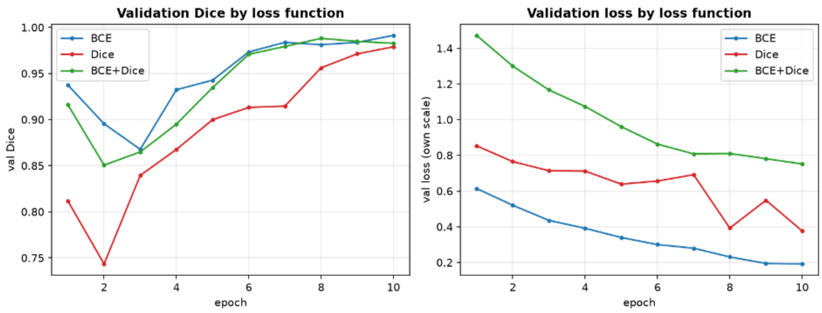


Figure 6: Validation Dice / loss curves

The U-Net mask was processed into a feature vector for each of the 12 unseen test images and passed to the identical numeric-only model used in Task 2. The results included a JSON record and narrative, which were stored in outputs/task4_hybrid_pipeline_records.csv. One sample entry is test_004, with 42 detections:

{"image_id": "test_004", "n_objects": 42, "mean_area": 339.5, "density_class": "dense", "quality_flag": "good"}

"The image contains a dense collection of objects, with most having an average area of around 340 pixels. The shapes are generally regular, but some irregularities were detected, indicating potential overlap or merging of the objects. Overall, the quality of the object detection appears to be good."

All three sparse test images (test_000, test_001, test_006, each containing 8 to 14 objects) were labelled with a quality flag of “poor,” with the explanation that they were “poor due to the presence of sparse objects.” That treats low object count as a sign of bad image quality, even though the numbers never suggested anything was wrong with the images themselves. When comparing predicted counts with the true counts, the U-Net shows a mean absolute error of **12.25** objects, with the largest mistakes occurring on the densest and most clustered cases. for example, **test_010** was undercounted by **42** objects (**78** vs **36**), and **test_005** by **34** (**49** vs **15**).

Otsu vs. U-Net, (Loss Comparison): The number of objects detected by Otsu and U-Net on the test set of 12 images is the same (mean absolute error 12.25 for both), representing two different approaches, trained and untrained and converging to the same result. It means that the undercount is a structural one and cannot be solved through better training, it happens in the case of touching objects becoming one connected component in a binary mask. In particular, it should be noted that the undercount appears even in the ground-truth mask (test_004: 75 nuclei, 44 components); it is corrected only by the use of the 16-bit instance labels that recognize touching nuclei.

Robustness: Blur equally affects both models (intersection-over-union (IoU **0.48** - **0.68**). On the other hand, low contrast negatively influences Otsu much more than U-Net: the former method does not significantly suffer from it (IoU **0.966** - **0.995**, when a threshold is selected relatively to each image), while the latter method tends to predict the entire image as foreground (IoU **0.02** to **0.22**), meaning its dependence on brightness pattern rather than scale sensitivity and relative segmentation.

***Foundation model**:* **MedSAM** (Ma et al., 2024), given Otsu's boxes as object proposals, scored a mean Dice of **0.881** against ground truth across the 12 test images, clearly behind the trained U-Net's **0.992** on the same images. Inspecting the individual masks shows MedSAM finding the same nuclei as the U-Net but with rounder, less precise edges. MedSAM was fitted on natural clinical scans: CT, MRI, endoscopy and not small synthetic fluorescence blobs; a general model did not beat one trained directly on this exact picture type.

6. Answering the five questions

1. **Which description is more useful, which is more trustworthy? VLM** is more useful when it works (run 2 named modality/subject correctly, unprompted). **Numbers-first** is more trustworthy: it can't invent a visual detail, only misjudge a given one. VLM run 1 called a sharp image "blurry" which was wrong, and only checkable against the image itself. A numbers-first error ("irregular" shape) checks against solidity 0.92 in one line.

2. **Did the U-Net improve on Otsu?** One example each. **Counting:** no, identical counts on all 12 test images. **Pixel quality:** yes, **val_019** (20 nuclei) scores Dice **0.996** with clean, separated shapes where Otsu typically leaves fusion. **Under corruption:** Otsu wins, **test_004_blur** gives Otsu **IoU 0.68** (20 regions) vs **U-Net's 0.48** (9, fused into blobs).

3. **Report your U-Net's Dice/IoU, meaning, and mistakes:** Final Validation Dice is **0.9929,** IoU **0.9859**, approx. 98.6% pixel agreement; every validation image scores above 0.988, no weak case. Real mistakes lie elsewhere: counting on touching nuclei (up to 42 short on the densest test image) and total collapse under low-contrast input never seen in training.

4. **Where in the pipeline can the LLM hallucinate, and does JSON help?** At both LLM steps: Task 1 stated a wrong fact confidently ("blurry"); Task 2 contradicted its own JSON against its own sentence. JSON and numbers-only input reduce invented visual detail but don't guarantee correctness, JSON constrains shape, not truth. Its real value is detectability: a broken parse returns None instead of passing bad text on, and quality_flag: "poor" beside n_objects: 15 is a one-line check a buried paragraph claim isn't.

5. **Considering accuracy, auditability, and the limits of your dataset, would you trust any part of this system in a real clinical setting?** Not a single piece of this system can be trusted to run automatically in a real clinical setting without human intervention. MedSAM lost to a model trained on this exact synthetic distribution, and the U-Net collapses outside it entirely. Otsu and raw JSON are most auditable, tracing to a formula or table. Best fix: an automatic check validating incoming images against the training histogram and every JSON response against successful parsing before either result is used.

References

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